

ANSWERS

1.

Question	Answer	Marks
5(a)(i)	infrared (radiation) and visible (light) and ultraviolet (radiation)	B1
	infrared (radiation) and visible (light) and ultraviolet (radiation) and in this order	B1
5(a)(ii)	damaging effect: (skin) cancer / cataracts	B1
	property: it / ultraviolet radiation is ionising	B1
5(b)	in a transverse wave, the <u>vibration</u> direction is perpendicular / at right angles to the propagation direction	B1
	in a longitudinal wave, the <u>vibration</u> direction is parallel to the propagation direction	B1

2.

Question	Answer	Marks
7(a)(i)	X-rays and gamma (radiation / rays)	B1
7(a)(ii)	security marking / detecting counterfeit bank notes / sterilising water / fluorescent effects (e.g. fingerprints, blood, biological fluids) / (sun) tanning / sun beds / cancer treatment / curing inks and resins (e.g. in dental fillings) / production of vitamin (D) / generating electricity / solar panels / disinfection / sterilisation	B1
7(a)(iii)	(skin) cancer / cataracts / burns or destroys the retina or cornea	B1
7(b)(i)	critical angle	B1
7(b)(ii)	$(n =) \sin i / \sin r$ or $1 / \sin c$	C1
	1.6	A1

3.

Question	Answer	Marks
4(a)	electromagnetic (waves) or components of electromagnetic spectrum	B1
4(b)(i)	number of oscillations / cycles / wavelengths	B1
	per unit time	B1
4(b)(ii)	gamma rays	B1
4(c)(i)	X-rays directed towards body part with (suspected) broken bone	B1
	X-rays pass through flesh but not (to same extent) through bones	B1
	image produced by digital / electronic / CCD / CMOS detector	B1
4(c)(ii)	X-rays (may) cause cancer / cell mutation / birth defect / miscarriage or X-rays show the bone structure rather than foetal development or image would not be clear	B1

4.

Question	Answer	Marks
5(a)	microwaves → satellite television	B1
	ultraviolet → sunbeds	B1
5(b)(i)	(microwaves) must pass through or not be absorbed (by container) or not reflected	B1

5.

Question	Answer	Marks
4(a)	(A:) ultraviolet (radiation) (B:) infra-red (radiation) (C:) microwaves any two correct	C1
	all three correct	A1
4(b)	gamma-rays and gamma-rays	B1
4(c)	X-rays pass through flesh and not (to the same extent) through bone	B1
	X-rays detected photographically or (digital) detector behind bone	B1
	no / less exposure / detection reveals bone or exposure / detection reveals break	B1

6.

Question	Answer	Marks
4(a)	$1.15 - 1.25 \times 10^{-6}$ (m)	B1
4(b)	infra-red	B1
4(c)	($f = v / \lambda$) algebraic or numerical	C1
	$3 \times 10^8 / 1.2 \times 10^{-6}$	C1
	2.5×10^{14} Hz	A1

7.

Question	Answer	Marks
10(a)	P - gamma(-rays) or γ (-rays)	
	Q - ultraviolet (radiation)	
	R - microwaves	
	any one correct	C1
	all three correct	A1
10(b)	P and X-rays and Q ticked	B1
10(c)(i)	($f =)c / \lambda$ or $3.0 \times 10^8 / 9.4 \times 10^{-7}$	C1
	3.2×10^N	C1
	3.2×10^{14} Hz	A1
10(c)(ii)	infra-red / radiation / signal / wave emitted by control and received at set	B1
	infra-red / radiation / signal / wave is encoded or is decoded	B1

8.

(a) (i) red

(ii) blue

(b) ANY 2 from (the use must agree with the type)

Microwaves

B1

use – satellite television, telephone, mobile / cell phones;
cooking, heating in a microwave oven, television
remote, radar, communication

B1

X(-rays)

B1

use – hospital use in medical imaging or security imaging, killing cancerous cells,
and engineering applications such as detecting cracks in metal, crystallography

B1

gamma (rays)

B1

use – medical treatment in killing cancerous cells, and engineering applications
such as detecting cracks in metal, sterilisation, tracer applications, radiotherapy

B1

9.

(a) gamma rays, visible light, infra-red

B1

(b) (i) microwaves

B1

(ii) satellite (receives and) sends/transmits/emits/boosts/amplifies signal

B1

(iii) cover a large area over the horizon / only one (transmitter/station) needed etc.
or unaffected by tall buildings/hills
or no obstructions

B1

10.

- (a) any **three** of:
infra-red and microwaves reversed
visible light is omitted
ultrasound is not e.m./should not be included
ultraviolet is missing ('ultrasound instead of light' scores 2) B3

- | | | | |
|---------------------------|----|--|----|
| (b) engineering use | M1 | detail/explanation | A1 |
| detecting cracks in metal | | (more) X-rays pass through crack/poor weld | |
| or | | or | |
| checking welds | | image of crack on film/screen | |
| astronomy | | hot stars emit X-rays | |
| crystallography | | diffraction reveals pattern of atoms | |
| fluorescence | | substances re-emit different energies | |
| (airport/border) security | | contents of luggage/lorries revealed | |
| paintings investigated | | underpainting revealed | |
| (not medical use) | | | |

11.

- (a) (i) X-ray(s) B1

- (ii) infra-red B1

- (b) any TWO lines:
same speed (in vacuo)
travel in a vacuum; need no medium
carry energy
transverse
can reflect/refract/diffract/interfere/polarise B2

- (c) microwaves B1

- 12.
- (a) $3(.00) \times 10^8 \text{ m/s}$ B1
- (b) 0.16 m **or** 16 cm B1
- (c) any **three** of:
travel through space/vacuum
pass through the atmosphere/not reflected by ionosphere
encoded (with the signal)
(satellite) amplifies/boosts signal
sent to/received by satellite
transmitted/sent by satellite
transmitted/received by a (satellite) **dish** (on Earth) B3
- (d) any **two** of:
same (high) speed (in air) **or** travel at speed of light
travel in vacuum/space **or** no medium needed
transfer/transmit energy
transverse (stated **or** explained)
(oscillating) magnetic **and** electric fields/waves
reflection/refraction/diffraction/interference/polarisation B2
- 13.
- (a) infrared B1
gamma (rays/waves) B1
- (b) (i) fluorescent (screen), photographic (plate), CCD/semiconductor/photoelectric/GM tube B1
- (ii) (X-rays) absorbed/stopped by bone **or** do not penetrate bone (**not** reflected by bone) B1
less absorption/pass **through** flesh/skin/body, etc. **or** travel in straight lines
or effect on detector, e.g. ionisation, photo black (on development), light emitted B1
- 14.
- (a) X-rays, ultra-violet, infra-red, microwaves in each box B2
allow one mark if moving one box gives correct order
- (b) sun-beds (accept tanning), fluorescent tubes, sterilisation, B1
illuminating marks on property (phosphors) **not** just marking property
- (c) transverse, same speed, will diffract, reflect, refract etc. (allow only 1) B2
travel in a vacuum (accept need no medium) any 2